Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 5.1 · A table

A survey of 100 users of two transport services A and B. Of these, 40 use
A, 35 use B and 15 use both.
the same probability of being chosen. A and B meaning respectively "user uses the
service A" and "user uses service B".

1. Build and complete a dual input table: whether or not to use A in lines, use
   or not of B in columns. Add totals.

2. Calculate $P_A(B)$ and $P_B(A)$ For each calculation, specify the reference population used in
   denominator.

3. Interpret both probabilities with two separate sentences.

Fiscal year 5.2 · Production

A company manufactures objects in two workshops A and B, which provide respectively $\frac{1}{4}$ and $\frac{3}{4}$ of
Of the items produced by A, 4 percent are defective; among those produced by A, 4 percent are defective.
by B, 2 percent are. One randomly selects an object from the whole production.
D The event "the object is defective".

1. Represent the situation by a weighted tree and specify the meaning of each probability of the second
   level.

2. Calculate $P(D)$ and interpret the result as an overall default rate.

3. We now know that the chosen object is defective. What is the probability that it was manufactured?
   In Workshop A?

Exercise 5.3 · Independence to be verified

We are launching a balanced dice with six faces numbered from 1 to 6. A : "Provide
an even number" and B : "To obtain a number greater than or equal to 4".

1. Write out the outcome sets corresponding to A, B and $A \cap B$.

2. Determine whether events A and B are independent. Justify using a numerical criterion.

3. Explain why they are not incompatible either.

Fiscal year 5.4 · No remission

One urn contains three red balls and two blue balls, indistinguishable to the touch.
chance a first ball, then a second without putting back the first. At each draw, all the
balls present in the urn are equiprobable.

1. Build the weighted tree of the two prints, distinguishing the color obtained from the first and the second second draw.
2. Calculate the probability of getting two red balls.
3. Calculate the probability of getting one ball of each color. Explain why two paths of the tree must be taken into account.

Exercise 5.5 · Three tests

An experiment involves a success of probability $\frac{1}{3}$ and a failure. We repeat it three times in a way independent, under the same conditions.

1. Build a tree on three levels, or list eight possible outcomes.
2. Calculate the probability of achieving exactly a success. Indicate the paths chosen.
3. Calculate the probability of achieving at least two successes by distinguishing the cases of two and three success.

Exercise 5.6 · Detector

A device controls objects of which 2 percent are defective. It triggers a signal for 90
per cent of defective objects and, wrongly, 5 per cent of flawless objects.
It's a random, controlled object. D the event "the object is defective" and S the event « device triggers a signal".

1. Determine the proportion of objects actually defective among the objects reported. representation and detailed reasoning.
2. Give an exact value and then a percentage rounded to the tenth.
3. A person directly offers 90 percent as an answer. Explain precisely what confusion of probabilities it commits.

Exercise 5.7 · Together and Cardinal

We all work together. $E = \{1; 2; 3; 4\}$. Its parts are considered $A = \{1; 2\}$ and $B = \{2; 4\}$.

1. Write $A \cap B$, $A \cup B$ and the supplement to A in E. Give each one's cardinal.
2. Write all couples of $A \times B$then give his cardinal.
3. Explain why couples (1; 2) and (2; 1) cannot be confused. to $A \times B$?

Fiscal year 5.8 · Deferred review

In a random experiment, two events A and B check

$$P(A) = \frac{1}{2}, \qquad P_A(B) = \frac{1}{4}, \qquad P_{\overline{A}}(B) = \frac{3}{4}.$$

1. Calculate $P(B)$ using a tree or total probability formula.

2. Calculate $P_B(A)$ by showing the common event used in the numerator.

3. Probability $P_A(B)$ and $P_B(A)$ are they equal here ? Is this equality a general property ?

Personal work. Perform only the task assigned at the end of the meeting, in progress Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions Resuming exercise 8 a week after the chapter (10 min). is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 67 p. 336, Questions 1, 2 and 3. Building a tree by keeping the D and N events in the manual. Attention: N is a negative test.

CQFD, exercise 70 p. 336, questions a and b. Count the four equiprobable outcomes of the two launches to refute the proposed equalitys.